

An Advanced Analysis System for Identifying Alcoholic Brain State Through EEG Signals

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Abstract: This paper addresses an advanced analysis system for the identification of alcoholic brain states from electroencephalogram (EEG) data in an automatic way. This study introduces an optimum allocation based sampling (OAS) scheme to discover the most favourable representative data points from every single time-window of each EEG signal considering the minimal variability of the observations. Combining all representative samples of each time-window in a set, some statistical features are extracted from every set of each class. The Mann-Whitney U test is used to assess whether each of the features is significant between the two classes (e.g., alcoholic and control). In order to evaluate the effectiveness of the OAS-based features, four well-known machine learning methods (decision table, support vector machine (SVM), k -nearest neighbor (k -NN) and logistic regression) are considered for identification of alcoholic brain state. The experimental results on the UCI KDD (i.e., UCI knowledge discovery in databases) database demonstrate that the OAS based decision table algorithm yields the highest accuracy of 99.58% with a low false alarm rate 0.40%, which is an improvement of up to 9.58% over the existing algorithms. A proposed analysis system can be used to detect alcoholism and also to determine the level of alcoholism-related changes in EEG signals.

Keywords: Electroencephalogram (EEG), alcoholism, optimum allocation technique, feature extraction, decision table.

1 Introduction

Alcoholism is one of the most common psychiatric disorders associated with considerable morbidity and mortality^[1]. According to the World Health Organization (WHO) report in 2014, almost 3.3 million people (5.9%) of deaths worldwide are due to alcohol consumption (WHO, 2014)^[2], which is the fifth leading cause of deaths^[3] and is the main risk factor for early death and disability^[4]. There are a wide range of effect of alcoholism in health such as, liver diseases, heart diseases, brain damage and certain cancers, etc. It is also a significant cause of other harms, such as road and other accidents, domestic, and public violence, crime, and contributes to family breakdown and broader social dysfunction (Ministerial Council on Drug Strategy (MCDS) 2011)^[5]. Alcoholics undergo numerous cognitive deficiencies, for instance, learning and memory deficits, impairment of decision making, and problems with motor skills, as well as suffering behavioural changes that include anxiety and depression^[6, 7]. The electroencephalogram (EEG) is one of

most important tools for studying brain events, functions, and disorders. EEG signals present recorded electrical activity produced by the firing of neuron within the brain along the scalp^[8]. Generally, recorded EEG signals are very complex in nature and contribute to a large amount of data to be analyzed. Understanding and interpreting the dynamics of the brain based on the EEG signals is very challenging for multi-channel sensor information. In real situations, visual inspection is the only way to detect dissimilarities in EEG signals by very skilled clinicians whether the signals come from healthy or alcoholic subjects. Even expert clinicians can miss the fine variations of the signals due to the existence of noise^[9]. Thus, the motivation of this study is to develop an automatic analysis system for the diagnosis of alcoholism with acceptable accuracy, due to the increased need for proper diagnosis and classification of neurological abnormalities. It will help us to have effective treatments and early warnings about the forthcoming diseases.

In the last few years, many methods have been developed in the literature for identifying alcoholism through EEG signals. In [10], a method based on approximate entropy (ApEn) features was proposed to discriminate between alcoholic and control EEG signals. Kannathal et al.^[11] used various features such as correlation dimension (CD), largest Lyapunov exponent (LLE), entropies,

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and Hurst exponent (H) to extract representative characteristics from the alcoholic and normal EEG signals. Acharya et al.^[12] developed an algorithm for the classification of alcoholic and control EEG data based on several nonlinear features such as, ApEn, LLE, sample entropy (SampEn), and four other higher-order spectra (HOS) features and a support vector machine (SVM) classifier. The change of EEG signals in the power distribution was measured using different frequency domain parameters such as, fast Fourier transform (FFT), autoregressive (AR) method, and performed classification task through the receiver operating characteristic (ROC) curve^[13]. In [14], a computer-aided analysis method was proposed for diagnosing alcoholic EEG signals using the second order AR model and principal component analysis (PCA) with an adaptive neuro-fuzzy inference system. Wavelet transformation (WT) based features were reported for the classification of alcoholic and control EEG signals by [15]. Frequency analysis, absolute and relative powers of the four classical bands were considered to determine alternation in alcoholic patients in [16]. In [17], a spectral entropy based algorithm was introduced for detection of alcoholic EEG signals. That study suggests using gamma band for discovering alcoholic EEG signals. A horizontal visibility graph entropy (HVGE) approach was reported in [18] for identifying alcoholic and controlled subjects and also compared with the SampEn method. In spite of numerous research works, the obtained results are not yet mature enough for automatic identification of alcoholic EEG signals and limited in their success with effectiveness. A challenging task to be tackled is to provide a good trade-off between high classification accuracy and a low false alarm rate.

Addressing the problems, this study aims to introduce an advanced analysis system, which considers the variability of the observations within a time-window improving the classification accuracy with a low false alarm rate for the recognition of alcoholic EEG pattern. In the alcoholic research area, none of the previous work considered the variability of the observations within a time-window although the variation of the observations with time is a vital concern for describing the characteristics of the raw signals. Thus, the present study proposes an optimum allocation based sampling (OAS) technique, which is very efficient in obtaining representative samples from every time-window considering the variability of the observations of each category of EEG data that reflects the entire data of the time-windows. In this proposed scheme, at first, we divide all the EEG signals in each class into several time-windows (called here “segments”) based on a particular time period. Then, we apply the OAS scheme to have representative data points from each segment of each category of EEG data. After that, we combine all of the selected samples from the segments in each category into one set named the “OAS” set and then statistical features are extracted from the “OAS” set in each class. We use the Mann-Whitney U test to evaluate whether

each of the features are significant between the two classes (e.g., alcoholic and control). After that, we employ four modern machine learning classifiers: decision table, support vector machine (SVM), k -nearest neighbour (k -NN) and logistic regression on the obtained feature vector set to assess the performance of the proposed methods. The various kernel functions of the SVM are tested to show their performance. In order to further verify the effectiveness of the proposed algorithm, the proposed method is compared with reported methods in the literature on the same dataset.

The rest of the paper is organized as follows. Section 2 presents a description of the proposed methodology and also provides a brief description about the experimental dataset. Section 3 provides some discussion about experimental setup. In Sections 4 and 5, we present the experimental results with detailed discussions. Finally, concluding remarks are included in Section 6.

2 Materials and methods

2.1 Dataset

The EEG data are obtained from human brains of alcoholic and control subjects (EEG database-UCI KDD (i.e., UCI knowledge discovery in databases) archive)^[19] for this study. It provides the 64 electrodes recoding on the scalps of the subjects. The positions of the electrode were located at standard sites (American Electroencephalographic Association 1990). The sampling frequency of recorded of the EEG signal is 256 Hz. There are two groups of subjects: alcoholic and control. In both groups, there are 122 subjects and each subject completed 120 trials. Each of the trails has 2048 data points recorded for 8 s. There are three versions of the EEG data set: the small data set, the large data set, and the full data set. In this study, The small data set is used for experimental purposes. Fig.1 shows the typical alcoholic and control

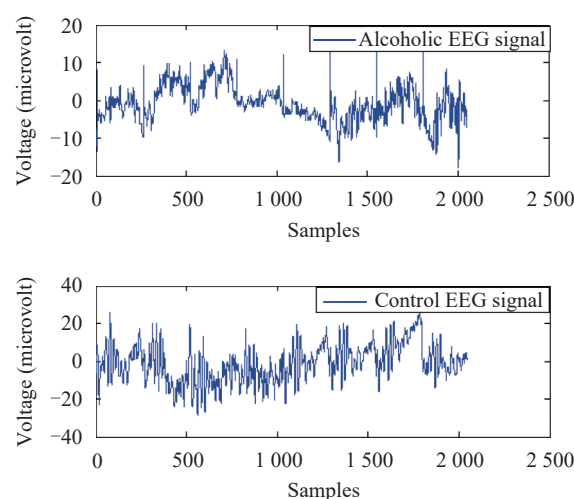


Fig. 1 Exemplary EEG signals for alcoholic and control subject

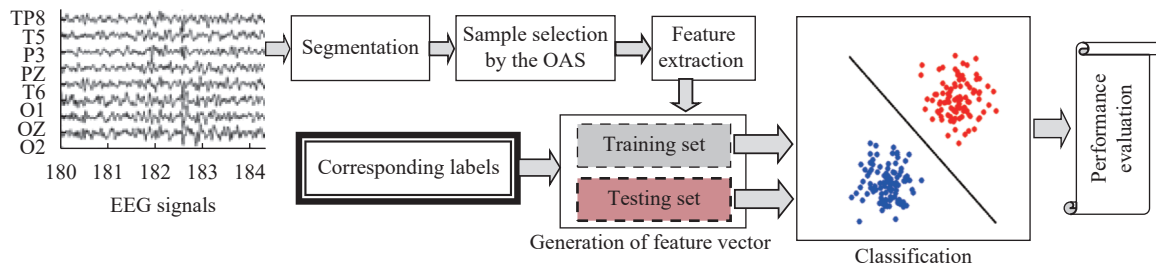


Fig. 2 Diagram of the overall system's architecture for identifying alcoholic EEG signals

EEG signals from an alcohol dependent subject and a healthy control subject.

2.2 Methods

In this study, an optimum allocation based sampling (OAS) scheme for classifying alcoholic and control EEG signals is introduced. The entire process of this method is divided into several processing modules: segmentation, sample selection by optimum allocation based sampling (OAS) technique, feature extraction, generation of feature vector set, classification and performance evaluation as shown in Fig. 2. The aforementioned modules are discussed below.

2.3 Segmentation

EEG signals are non-stationary, aperiodic in nature and the magnitudes are changed with respect to time. However, in the signal analysis, it is required to make an EEG signal stationary. Though a whole EEG signal may not be stationary, the segments exhibit stationarity properties. In this step, EEG signals are partitioned into some time-windows called "Segment" considering a specific time period based on data structure (please see an example in Fig. 3). These segments are mutually exclusive groups. Thus, the purpose of segmentation is to properly account for possible stationarities in the EEG signals because most of the signal analysis techniques require the signals to be static. In this method, EEG signals are divided into some mutually exclusive k segments such as, $Segment_1, Segment_2, \dots, Segment_k$, considering a particular time period to have representative values of specific time period. Suppose here, the number of observations

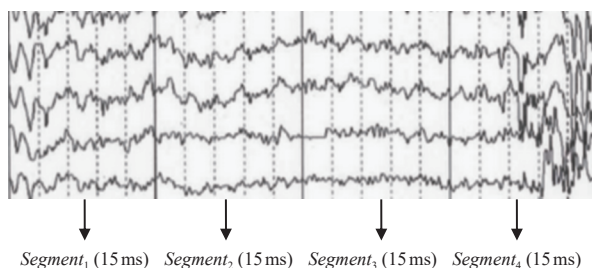


Fig. 3 An illustration of segmentation for both alcoholic and control EEG signals

in $Segment_1, \dots, Segment_k$ are $N_1, N_2, \dots, N_k$, respectively. It is worth noting that, for any experiment, the number of segments (k) is determined empirically over time based on the data size. Fig. 3 presents an example of EEG signals which are partitioned in specific time period. Here, each segment contains 15 ms (millisecond) data while the whole signals consist of 1 s data. In this example, k is considered as 4. After segmentation, the representative observations from each segment are selected by using the OAS technique.

2.4 Sample selection by the OAS scheme

This step aims to select a representative sample from every segment of a category of EEG signals considering a minimum variance. Here, "a representative sample" refers to a representative part (or subset) of a segment that reflects the whole EEG data of that segment. An observation in a sample is called the sample unit, and the sample size is the number of observations that are included in a sample. Generally, in a random sample section, variability is not considered within a group which is most important thing to provide the precision of the sample. Optimum allocation based sampling is used to determine the number of observations to be selected from different segments considering variability among the values. If the variability within a segment is large, the size of a sample from that segment is also large. On the other hand, if the variability of the observations within a segment is small, the sample size will be small in that segment. The required sample from each segment using the OAS scheme is obtained by (1) considering the variability among the signals in each segment. A detailed description of the OAS is available in [20].

$$n(i) = \frac{N_i \sqrt{\sum_{j=1}^p \sigma_{ij}^2}}{\sum_{i=1}^k \left(N_i \sqrt{\sum_{j=1}^p \sigma_{ij}^2} \right)} \times m$$

$$i(\text{no. of segment}) = 1, 2, \dots, k$$

$$j(\text{no. of signal}) = 1, 2, \dots, p \quad (3)$$

where $n(i)$ is the required sample size of the i -th Segment; N_i is the data size of the i -th Segment; σ_{ij}^2 is

the variance of the j -th signal of the j -th Segment; and m is the total sample size of the EEG recording of a class is obtained by (2)^[21]

$$m = \frac{n_0}{1 + \frac{n_0 - 1}{N}} \quad (4)$$

where N refers to the number of observations in a class and n_0 is the preliminary sample size calculated by $n_0 = \frac{z^2 \times p \times (1 - p)}{e^2}$. If the estimator p is not known, 0.50 (50%) is used as it produces the largest sample size. In this research, we consider $p = 0.50$ so that the sample size is maximum and $Z = 2.58$ and $e = 0.01$ for 99% confidence level for calculation of n_0 .

If $n(1), n(2), \dots, n(k)$ are the calculated sample sizes obtained by (1) from $Segment_1, Segment_2, \dots, Segment_k$, respectively, we select the required sample (a representative subset) from each segment considering these sample sizes. Then, all of the selected samples from the segments of each class are combined in a set, named "OAS set" and after that, the representative characteristics are extracted from the "OAS set" as discussed in the following step.

2.5 Feature extraction

In this study, the considered features are mean, maximum, minimum, standard deviation, median, mode, first quartile (Q_1), third quartile (Q_3), inter-quartile-range (IQR), kurtosis and skewness, which are based on statistical measures of the input EEG signals. These features are selected because of their efficiency in representation of the EEG signal in each segment. In practice, EEG data can be symmetric or skewed. Because of this fact, a symmetric distribution which can appropriately measure the centre and variability of the data by employing the mean and standard deviation is needed. In addition, median and IQR values can be employed to measure the centre and spread of the data in the sense of skewed distributions^[22]. The mode feature for a given continuous probability distribution is defined as the peak of its histogram or density function. Skewness describes the shape of a distribution that characterizes the degree of asymmetry of a distribution around its mean^[22]. Kurtosis measures of whether the data are peaked or flat relative to a normal distribution. These features are extracted from the "OAS set".

2.6 Generation of feature vector set

After feature extraction, all of the features extracted from the alcoholic and control group are combined in a feature vector set generating one feature space. Finally, the acquired features incorporate spatial information from the multi-channel EEG data to obtain maximum accu-

acy and efficiency for the classification of alcoholic and control EEG patterns. The efficiency of the extracted features is evaluated by using 10-fold cross-validation in which the feature vector is divided into training and testing sets.

2.7 Classification and performance evaluation

This step focuses on the classification process of the alcoholic and control EEG signals employed on the obtained feature vector set to indicate the general direction of the data set. The tested four modern machine learning methods: Decision tree^[23], support vector machine (SVM)^[24, 25], k -nearest neighbor (k -NN)^[26, 27] and logistic regression^[28–30] to assess the performance for the obtained feature vector set. As per my knowledge, we introduced the decision table classifier for first time in this research area for identifying an alcoholic EEG pattern. After classification, the performances are evaluated by accuracy^[31–33], true positive rate (TPR) or sensitivity, false alarm rate (FAR) or false positive rate (or 1-specificity), precision, F-measure, kappa statistic and receiver operating characteristic curve (ROC) area^[34].

3 Experimental setup

With the purpose of assessing the performance and consistency of the proposed method, we implement our analysis scheme on the UCI KDD database for identifying alcoholic brain states through multi-channel EEG signals. Following the proposed methodology design, firstly we segmented each of two classes (e.g., alcoholic and control group) into four parts according to 15ms as a particular time period based on the data structure. As every signal in each of two classes contains 2048 observations of 1s, the sizes of the four segments in each class are $Segment_1, N_1 = 512$; $Segment_2, N_2 = 512$; $Segment_3, N_3 = 512$; and $Segment_4, N_4 = 512$, respectively, and each segment contains the data for 15ms. Secondly, we selected a sample (a representative subset of a segment) randomly from each of four segments in every class using the OAS technique as discussed in Section 2.4. In the OAS scheme, (1) was used to compute the sample sizes for each segment considering $N = 2048$ and $m = 1824$ in both alcoholic and control group where N refers to total number of observations in each class and m refers to overall sample size in each class discussed in Section 2.3. An example of calculation procedure of the proposed method on the UCI KDD database is shown in Fig. 4.

The obtained sample sizes for each segment by the OAS technique are reported in Table 1 according to classes. On the basis of the obtained sample sizes (e.g., $n(1) = 425$), the samples were selected from the respective segment of that class. We combined all of the samples obtained from each of four segments for an individual

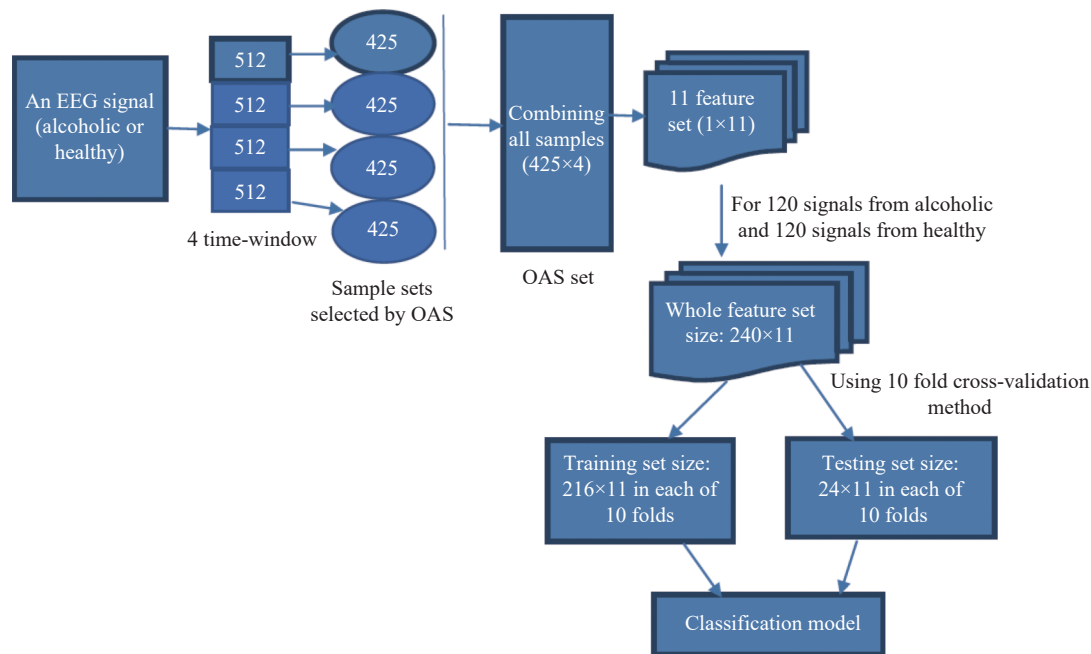


Fig. 4 An illustration of implementation process of the proposed scheme on the UCI KDD database

Table 1 Calculated sizes of sample for each of four segment

Classes	Sizes of sample by the OAS technique								Total
	<i>Segment</i> ₁		<i>Segment</i> ₂		<i>Segment</i> ₃		<i>Segment</i> ₄		
	<i>N</i> ₁	<i>n</i> (1)	<i>N</i> ₂	<i>n</i> (2)	<i>N</i> ₃	<i>n</i> (3)	<i>N</i> ₄	<i>n</i> (4)	
Alcoholic category	512	425	512	455	512	390	512	512	<i>n</i> = 1782, <i>N</i> = 2048
Control category	512	446	512	447	512	405	512	512	<i>n</i> = 1810, <i>N</i> = 2048

Note: N_1 = sample size of 1st *Segment*; $n(1)$ = calculated sample size by the OAS for 1st *Segment*; $n = n_1 + n_2 + n_3 + n_4$

class called the “OAS set” of that class. The extracted eleven statistical features from the “OAS set” in each class represent the distribution pattern of that class. The reasons to considering the eleven features in this study are discussed in detail in Section 2.5. After that, we generated one feature vector set combining the obtained features from both classes. The size of the feature vector for each class is 120×11 and thus the size of the whole feature vector for two-class is 240×11 in this process. After that, 10-fold cross-validation process was employed to create a training set and testing set for performance evaluation of the proposed algorithm. In each of 10 iterations, the training set had 216×11 data point while the testing set had 24×11 data point.

In order to test the significance of each of the features between two groups/classes (e.g., alcoholic and control), we used a Mann-Whitney U test in this study. This test helps to assess whether the distributions of each feature are the same across categories of class. This check also helps to determine whether data groups have distinguishable characteristics or not, even if they show the same mean value. Clinically significant features will have lower “ p -value”, such as < 0.05 . In Table 2, the features

are listed in the first column. The second and third column contains the range (mean $\pm$ standard deviation) of each feature extracted from the alcoholic and control group, respectively. The last column holds the “ p -value” which was evaluated using the Mann-Whitney U test. The results show very low “ p -value” (< 0.05), indicating that the results are clinically significant. Table 2 shows that all features except “mode” are clinically significant (lower p -values).

Following the test results reported in Table 2, we performed experiments for our proposed method which considering mode (e.g., mean, max, min, std, median, mode, Q1, Q3, IQR, kurtosis and skewness) and which did not consider mode (e.g., mean, max, min, std, median, Q1, Q3, IQR, kurtosis and skewness) in the feature vector set. These two feature vector sets were individually fed into each of four machine learning methods: decision table, SVM, k -NN and logistic regression as inputs to test the performance of the two separate feature sets. We tested four classification algorithms: Decision table, SVM, k -NN and logistic regression, implemented in Waikato environment for knowledge analysis (WEKA) machine learning toolkit. LIBSVM was used for the SVM classific-

Table 2 Statistical significant test for the considered features

Features	Alcoholic	Control	p-value
Mean	$0.238\,9 \pm 2.375\,16$	$-0.666\,2 \pm 4.376\,13$	$0.017 < 0.02$
Maximum	$38.312\,1 \pm 15.871\,01$	$3.687\,2 \pm 12.650\,72$	$0.007 < 0.01$
Minimum	$-20.096\,3 \pm 7.927\,44$	$-34.305\,1 \pm 12.686\,16$	$0.000 < 0.01$
Standard deviation	$5.858\,8 \pm 1.386\,86$	$9.956\,6 \pm 1.686\,16$	$0.000 < 0.01$
Median	$0.123\,0 \pm 2.181\,66$	$-0.633\,0 \pm 4.074\,33$	$0.024 < 0.03$
Mode	$-0.132\,5 \pm 23.425\,61$	$3.101\,9 \pm 15.915\,17$	0.348
1st quartile (Q1)	$-3.034\,6 \pm 2.575\,43$	$-6.739\,8 \pm 4.471\,94$	$0.000 < 0.01$
3rd quartile (Q3)	$3.341\,6 \pm 2.325\,55$	$5.205\,9 \pm 4.412\,15$	$0.000 < 0.01$
Inter-quartile-range (IQR)	$6.376\,1 \pm 1.750\,07$	$11.945\,6 \pm 2.461\,47$	$0.000 < 0.01$
Kurtosis	$15.274\,2 \pm 16.428\,54$	$5.579\,5 \pm 3.535\,82$	$0.000 < 0.01$
Skewness	$1.204\,8 \pm 1.735\,71$	$0.283\,4 \pm 0.708\,36$	$0.000 < 0.01$

ation in WEKA. We considered the parameter values of the above mentioned classification methods in WEKA default parameters settings in this study.

4 Results

Table 3 presents the overall classification results of the proposed method with the eleven (with mode) and ten feature set (without mode). In Table 3, the overall performances for each classifier in terms of accuracy, true positive rate, false alarm rate, precision and F-measure are reported. As shown in Table 3, the overall accuracy for the decision table, SVM (polynomial kernel), k -NN and logistic regression classifiers with eleven feature set are 99.58%, 90.83%, 89.17% and 90.83%, respectively while these values for the ten feature set are 90.42%, 90.00%, 89.17%, 89.58%, respectively. Again, the overall true positive rates for the eleven feature set are obtained

99.60%, 90.80%, 89.20% and 90.80% by the decision table, SVM (polynomial kernel), k -NN and logistic regression classifiers, respectively while the values for the ten feature set are 90.40%, 90.00%, 89.20% and 89.60%. From the overall accuracy and true positive rate results, we can see that the eleven feature set can generate higher performance with each of the four proposed classifiers compared to the ten feature set and the decision table is the best classifier to produce the utmost performance with the eleven features set (accuracy, 99.58% and true positive rate, 99.60%).

The same properties are seen in the overall false alarm rates, overall precisions and overall F-values. The eleven feature set produces better results with the decision table classification method than the ten feature set in each of the performance evaluation measures. The decision table with eleven the feature set yields the lowest false alarm rate (0.40%) and the highest precision value (99.60%) and

Table 3 Results of the proposed method considering the two feature sets

Classification methods	Accuracy	True positive rate	False alarm rate	Precision	F-value
Overall performances (%) for the eleven features					
Decision table	99.58	99.60	0.40	99.60	99.60
SVM (poly kernel)	90.83	90.80	9.20	90.90	90.80
SVM_linear kernel	90.0	90.0	10.0	90.00	90.00
SVM_RBF kernel	70.83	70.80	29.20	79.90	68.40
k -NN	89.17	89.20	10.80	89.30	89.20
Logistic regression	90.83	90.80	9.20	90.80	90.80
Overall performances (%) for the ten feature					
Decision table	90.42	90.40	9.6	90.40	90.40
SVM (poly kernel)	90.00	90.00	10.00	90.00	90.00
SVM_linear kernel	89.58	89.60	10.40	89.60	89.60
SVM_RBF kernel	74.58	74.60	25.40	79.00	73.60
k -NN	89.17	89.20	10.80	89.30	89.20
Logistic regression	89.58	89.60	10.40	89.60	89.60

F-value (99.60%). However, the lowest false alarm rate (0.40%) and the highest precision value (99.60%) and F-value (99.60%) were obtained by the decision table classifier. It is worth mentioning that the lower false alarm score indicates a higher performance for a classification method. Thus, the overall false alarm rate, precision and F-value results also prove that the eleven feature set can generate higher performance with the decision table classifier. The comparison in Table 3 clearly shows the significant improvements in all performance evaluation criteria (e.g., accuracy, true positive rate, false alarm rate, precision and F-value) for the eleven feature vector set compared to the ten feature set. The decision table classifier with the eleven feature set consistently yields the highest performance in all cases. From the results, it is seen that owing to considering mode, the decision table can generate better performance. There are some important features in EEG signal analysis like mode although it is not showing significant in the assessment but providing significant impact in the classification performance. As in the most of the cases, EEG signals are asymmetric in shape, the mode can be vital to reflect the original signal pattern. Thus, for further assessment of our proposed methodology, we considered the eleven features set as an optimum feature set for the classification of alcoholic and control EEG data. It is worth mentioning that we tested the SVM classifier with four kernel functions such as, polynomial kernel, linear kernel, and radial basis function (RBF) kernel with the feature sets. In both cases, the SVM with polynomial kernel yields better performances than with the other two kernel functions in the identification of alcoholic EEG signal. Thus, we considered an SVM classifier with a polynomial kernel function in comparison with the other reported methods.

In order to further exhibit the effectiveness of the proposed method with the eleven feature set, we provide the performance information in both alcoholic and control groups in terms of class-specific true positive rate, false alarm rate, precision and F-values displayed in Figs. 5–8, respectively. The error bars in the graphs represent the standard errors indicating the fluctuations in the performances among the classifiers. Fig. 5 shows the comparison of true positive rates for the four classification methods in alcoholic and control group. The dissimilarity of false alarm rates for the four classification methods in alcoholic and control groups are presented in Fig. 6. As this shows, the lowest false alarm rate is obtained in both categories for the decision table classification method among the other three methods. Fig. 7 illustrates the variation of the precision for the four classifiers in two categories. The difference of the F-values for the reported classifiers in alcoholic and control groups are displayed in Fig. 8. As shown in Figs. 5–8, among the reported four classifiers, the decision table classifier produces the best performances when using eleven features in each category, while the k -NN classifier consistently displays the lowest per-

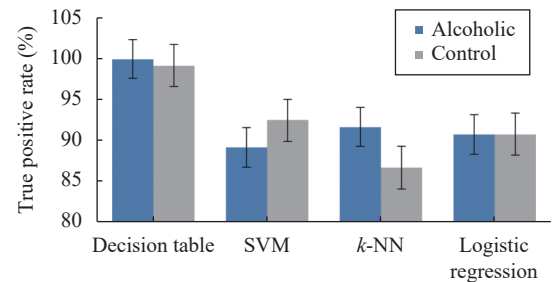


Fig. 5 True positive rates in alcoholic and control group for the four proposed classifiers

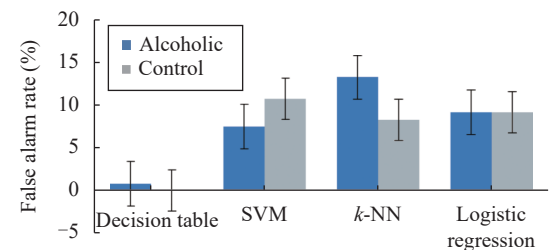


Fig. 6 False alarm rate for the four proposed classifiers in alcoholic and control group

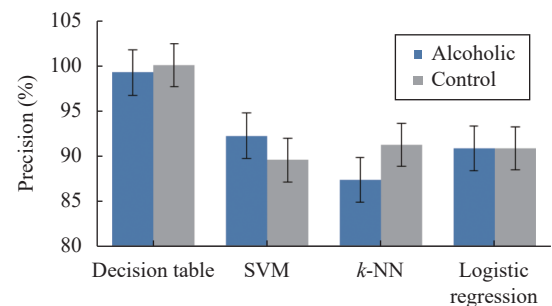


Fig. 7 Precision for the four proposed classifiers in alcoholic and control group

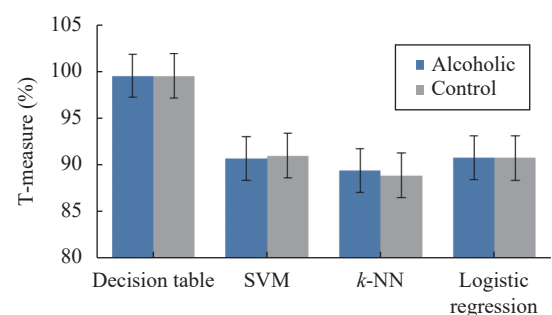


Fig. 8 False alarm rate for the four proposed classifiers in alcoholic and control group

formances.

In order to show more information about the performance of the proposed method with the eleven feature set, the values of the area under ROC and kappa statistics are presented in Figs. 9(a) and 9(b), respectively. The area under a ROC curve measures the overall capability

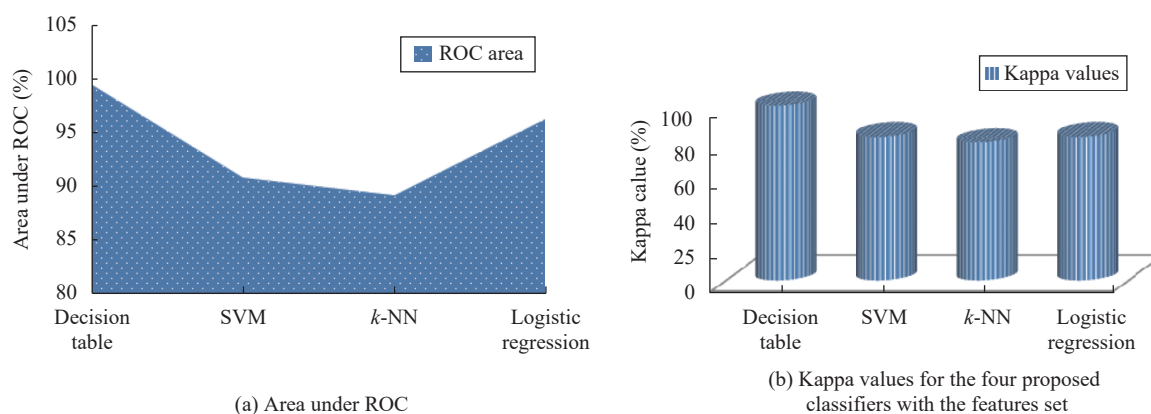


Fig. 9 Evaluation of performances

to discriminate between alcoholic and control groups. The values of area under ROC lie between 0–1 (0%–100%). It will be truly a useless test, if it is $\leq 50\%$; poor if it lies between 60%–70%; fair if it lies between 70%–80%; good if it is between 80%–90%; and excellent if it is between 90%–100%. As shown in Fig. 9(a), the value of the ROC area is close to 99.40% for the decision table, 90.80% for the SVM, 89.20% for the *k*-NN and 96.20% for the logistic regression. The results demonstrate that the decision table has the highest ability to discriminate between alcoholic and control EEG signals using the eleven feature set among the other four classification methods. Fig. 9(b) displays the kappa value for all reported classifiers assuming the eleven features as input. The aim of the kappa statistics test is to evaluate the consistency of the classifiers. Consistency is considered mild if kappa values are less than 20%, fair if it lies between 21%–40%, moderate if it lies between 41%–60%, good if it is between 61%–80%, and excellent if it is greater than 81%. As shown in Fig. 9(b), the highest kappa value (99.17%) is obtained by the decision table method while the values for other three classifiers: SVM, *k*-NN and logistic regression methods are 81.67%, 78.33% and 81.67%, respectively. Once again the decision table classifier seems to be the best choice to classify the alcoholic EEG signals from control signals. At the end, based on our experimental results, we can say that the proposed method has the ability to produce higher performance which may assist in making advances in computer-aided assessment and also to conduct an automated neurophysiological assessment for the detection of changes associated with alcoholism.

5 Discussion

There are a few studies in the literature (as discussed in the introduction section) on the identification of alcoholic EEG signals such as, [11–13, 18, 35–42], that were performed on the benchmark UCI KDD database. In order to further examine the efficiency of our proposed OAS based algorithm, a comparative study is presented between our proposed method and the five well-known

existing methods for the same database in Table 4. This table reports the performances of the existing methods in various measures as different research reported their outcome in different principles. The highest classification performances among the five algorithms are highlighted in bold font. From Table 4, it is clear that the proposed OAS algorithm with the decision table classifier yields the highest classification performances in each evaluation criteria among all reported existing methods in the literature. For example, the accuracy, sensitivity, specificity and ROC area of the proposed method is 99.58%, 99.60%, 99.60%, 0.9940 (99.40%), respectively while these values are maximum 98.10% by Zhu et al.^[18], and 95.80%, 95.80% and 0.8222 (82.22%), in previous research work, obtained by Faust et al.^[35]

The lowest accuracy (88.00%) was reported by Ehlers et al.^[38]. Very recently, the various methods have been proposed for classification of alcohol and control EEG signals^[40–42]. Thus, our proposed algorithm improves the accuracy by 1.48–11.58% compared to the reference methods. The results indicate that the OAS based approach can be used as a perfect scheme for feature extractions while the decision table can be considered as an optimum choice with it for the identification of alcoholic EEG signals. There are a few limitations of this study. Firstly, this paper focuses on a two-class classification problem (alcoholism/healthy), although we plan to extend the proposed algorithm to multiclass situations in the near future. Secondly, we tested our proposed method on a publically available data set for the purposes of comparison. In future, we have a plan to improve this proposed method for application in the detection of various neurological diseases such as dementia, brain tumours, sleep disorder, depression, etc.

6 Conclusions

This paper presents an advanced analysis system to identify alcoholic EEG signals using the OAS scheme adopting a suitable machine learning algorithm. The OAS technique is employed to select representative samples

Table 4 Classification performance comparison with recently presented same dataset methods

Authors	Method	Features	Classifier	Performances (%)
Acharya et al. ^[12]		Approximate entropy, sample entropy, Lyapunov exponent, higher order spectra	SVM with poly kernel	Accuracy = 91.70; Sensitivity = 90.00; Specific = 93.30
Faust et al. ^[32]	Wavelet packet decomposition	Relative energy	k-NN	Accuracy = 95.80; Sensitivity = 95.80; Specific = 95.80
Zhu et al. ^[18]	Horizontal visibility graph entropy (HVGE) approach		SVM with RBF kernel	Accuracy = 98.10
Kannathal et al. ^[11]		Correlation dimension, Lyapunov exponent, discriminant analysis, entropy, hurst's exponent	Unique ranges	Accuracy = 90.00
Faust et al. ^[13]	Burg's method	Peak amplitude and frequency		Area under ROC
Bajaj et al. ^[36]	Time-frequency	Features	Nonnegative least squares classifier (NNLS)	Accuracy = 95.83
Patidar et. al. ^[37]	Tunable Q-factor wavelet transform (TQWT)	Correntropy	Least square (LS)-SVM	Accuracy = 97.02
Ehlers et al. ^[38]		Correlation dimension	Discriminant analysis	Accuracy = 88
Faust et al. ^[39]	Higher order spectra (HOS)	Cumulants	Fuzzy Sugeno classifier (FSC)	Accuracy = 92.4
Taran and Bajaj ^[40]	Empirical mode decomposition (EMD) based rhythms	Statistical features	Extreme learning machine (ELM)	Accuracy = 97.92
Manish et al. ^[41]	Dual-tree complex wavelet transform		LS-SVM	Accuracy = 97.91
Priya et al. ^[42]	EMD	Statistical features	LS-SVM	Accuracy = 97.92
Proposed approach	Optimum allocation based sampling	Statistical features	Decision table	Accuracy = 99.58; True positive rate (sensitivity) = 99.60; False alarm rate specificity = 99.60; Area under ROC 0.9940 (99.40); Precision = 99.60; F-value = 99.60; Kappa statistics = 99.17

from every time-window (called “Segment”) considering the variability of the observations within a segment. The selected samples by the OAS from every segment in each category were combined with a subsequent statistical feature extraction process which is adopted with four different machine learning methods (decision table, SVM, k -NN and logistic regression) to evaluate the performance of the obtained features. A Mann-Whitney U test was used to assess whether each of the features was significant between the two classes (e.g., alcoholic and control). To assess the effectiveness of the proposed method, a 10-fold cross-validation method was used.

For further performance evaluation, the proposed algorithm was compared with five well-known existing methods. The experimental results show that a remarkably high accuracy rate (99.58%) with a low false alarm rate (0.40%) was achieved with the OAS based decision table approach. The results also demonstrate that our method is superior in comparison to the existing methods for the same alcoholic EEG database. This research leads us to confirm that the OAS scheme can be reliable

for capturing the valuable information from alcoholic EEG signals and the decision table is very promising with it for the identification of alcoholic brain states. This proposed approach can assist researchers or experts in developing automatic online computer-aided diagnosis systems for analyzing the alcoholic status of the human brain.

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